llmXive Automated Review of EVA-Bench: A New End-to-end Framework for Evaluating Voice Agents

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a robust evaluation framework, and most factual claims are well-supported by the data in Table 1. However, a few specific issues regarding citation accuracy and numerical precision require attention to ensure strict claim accuracy.

First, the Introduction cites `\cite{goodfeli/dlbook_notation}` (visible in the critical elements list) to support the claim that voice agents face "ephemeral speech, real-time timing, and variable acoustics." This citation points to a GitHub repository for LaTeX notation, which does not support the scientific claim. This must be replaced with a relevant primary source on speech processing or conversational dynamics, such as Stivers et al. (2009) or a similar foundational paper.

Second, the Introduction states that "Only \gptrealtime~(0.47, 0.57) clears 0.4 on both." Table 1 lists the EVA-X \passatone\ for \gptrealtimefull\ as `0.566`. While 0.566 rounds to 0.57, presenting the rounded value in the text without clarification creates a minor discrepancy with the table's precision. Given the paper's focus on metric thresholds, the text should either use the precise value (0.566) or explicitly note that values are rounded to two decimal places.

Third, the "Main Findings" section claims: "S2S systems dominate EVA-X (mean turn-taking 0.82–0.58 vs. cascade 0.28–0.58)." The range 0.28–0.58 for cascade systems corresponds to the minimum (0.283) and maximum (0.583) scores in Table 1. However, the text labels this as "mean turn-taking," which implies an aggregate calculation not shown in the table. To avoid ambiguity, the text should clarify that these figures represent the observed range of individual system scores rather than a calculated mean.

Finally, the claim regarding a "median \passatk–\passpowerk~gap of 0.44 on EVA-A" is a derived statistic. While likely correct based on the table data, the text does not explicitly show this calculation. For full transparency, the authors should ensure this derived metric is either explicitly calculated in the appendix or the text clarifies that this is a summary of the table data.

These are minor issues that do not undermine the paper's core findings but are necessary to ensure the factual claims are precisely supported by the cited evidence and data.

Action items.

- [**writing**] Replace the citation 'goodfeli/dlbook_notation' in the Introduction with a primary source on voice agent constraints (e.g., Stivers et al. 2009) as the current link is a LaTeX guide, not a scientific source for the claim.
- [**writing**] In the Introduction, clarify the rounding of the EVA-X score for gptrealtime. The text states 0.57 while Table 1 shows 0.566. Use the precise value or explicitly state the rounding convention to ensure factual precision.
- [**writing**] In 'Main Findings', clarify that the '0.28–0.58' range for cascade turn-taking refers to the observed range of individual system scores in Table 1, not a calculated mean, to ensure the claim is directly verifiable from the provided data.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 presents per-pipeline-type performance across four facets with error bars and individual system points, but the 'Violates policy' subplot mislabels its axis direction relative to the metric's meaning, and the legend lacks explicit symbol-to-type mapping despite relying on caption text for interpretation.

Figure 2

The figure is severely compromised by the complete absence of x-axis model labels, which are essential for interpreting the grouped bars described in the caption. Additionally, the caption contains a missing metric name, and the visual representation of the Hybrid section (grouped bars) contradicts the description of it being shown as individual points.

Figure 3

The figure is rendered without the necessary x-axis labels to identify the systems, and it lacks a legend to map the bar colors to the perturbation conditions described in the caption.

Figure 4

Figure 4 is critically flawed because the x-axis lacks model labels, preventing identification of the systems being compared. Additionally, the color legend is missing from the visual, and the caption contains a typo omitting the specific metric name.

Figure 5

Figure 5 is fundamentally broken because the x-axis labels are missing, preventing identification of the models being compared. Additionally, the caption omits the specific metric name, and there is a potential contradiction between the 'cascade systems' title and the caption's model list.

Figure 6

The figure displays perturbation effects but fails to identify the specific models on the x-axis or the metric on the y-axis, rendering the data uninterpretable without external context.

Figure 7

The figure is visually clear with labeled values and error bars, but the caption contains unrendered text placeholders for model names and contradicts the visual sorting order of the bars.

Figure 8

Figure 8 is a clear and well-structured pipeline diagram that effectively visualizes the framework overview described in the caption. All major components (Inputs, Simulation, Validation, Measurements) are clearly labeled with their respective sub-modules, and the flow of data and control logic is easy to follow.

Figure 9

The figure is visually clear with a functional legend, but the caption is severely broken with missing text and LaTeX artifacts that obscure the specific systems being discussed. Additionally, there is a contradiction between the legend's inclusion of a Hybrid mean point and the caption's description of Hybrid data presentation.

Figure 10

The figure is visually clear and the data presentation is consistent with the caption's description, but the caption text itself is truncated and contains raw LaTeX formatting codes.

Figure 11

The figure effectively visualizes the strong positive correlation between transcription accuracy and task completion for cascade systems, but the caption inaccurately describes the plot as showing 'correlation' (a single value) rather than the individual system data points shown, and the statistical summary is embedded in the image title rather than the caption.

Figure 12

The figure effectively visualizes the relationship between trial count and CI width, but the legend is too small to read, and the error bars at k=5 contradict the caption's claim that the width should be zero.

Action items.

- [**science**] Figure 1: The 'Violates policy' subplot (bottom-right) has an x-axis labeled 'Rate (lower is better)' but displays values near 0.5–0.8, which are high violation rates — yet the caption and plot imply these are good outcomes. This contradicts the metric definition and misleads interpretation; either the axis label or the plotted values are incorrect.
- [**writing**] Figure 1: The legend at the bottom uses colored symbols (circle, diamond, square, dot) but does not explicitly map them to pipeline types in the legend itself — it relies on the caption's text description ('Cascade', 'Hybrid', etc.), which is not visually linked to the symbols in the plot area. Add direct symbol-to-type labels in the legend for clarity.

- **[fatal]** Figure 2: The x-axis labels listing the specific models (e.g., 'Cascade: + + , ElevenAgents...') are completely missing from the rendered image, making it impossible to identify which bars correspond to which systems despite the caption's detailed list.
- **[science]** Figure 2: The caption states 'Hybrid ($n = 2$) is shown as individual system points only' (referencing Figure 1's style), but the Hybrid section in this figure displays grouped bars with error bars, contradicting the description of how Hybrid data is presented.
- **[writing]** Figure 2: The y-axis label 'Mean Delta' is present, but the caption text 'Perturbation effect on across all evaluated systems' contains a missing metric name (likely 'conversation progression' based on the filename), rendering the figure's specific target undefined.
- **[fatal]** Figure 3: The x-axis labels listing the specific models (e.g., 'Cascade: + + , ElevenAgents...') are completely missing from the rendered image, making it impossible to identify which bars correspond to which systems as described in the caption.
- **[science]** Figure 3: The caption states that bar colors encode perturbation conditions (accent, background noise, both), but there is no legend in the figure or caption defining which color corresponds to which condition.
- **[fatal]** Figure 4: The x-axis is completely missing labels; the caption lists specific models (e.g., 'Cascade: + + , ElevenAgents...') but the rendered figure has no text under the bars to identify which system corresponds to which data group.
- **[fatal]** Figure 4: The y-axis label 'Mean Delta' is present, but the caption text 'Perturbation effect on across all evaluated systems' contains a missing metric name (likely 'task completion' based on the filename), making the figure's specific subject ambiguous.
- **[science]** Figure 4: The caption states 'Bar colors encode the perturbation condition' (accent, bgnoise, both), but the legend defining which color maps to which condition is missing from the figure itself.
- **[fatal]** Figure 5: The x-axis is completely illegible; model names and categories are missing, making it impossible to identify which systems correspond to the bar groups.
- **[fatal]** Figure 5: The y-axis label 'Mean Delta' is present, but the caption text 'Perturbation effect on for cascade systems' is missing the specific metric name (e.g., 'transcription accuracy'), rendering the figure's subject ambiguous.
- **[science]** Figure 5: The caption lists 'Hybrid' and 'S2S' models, but the figure title claims to show 'cascade systems' only; the x-axis labels (if visible) would be needed to verify if non-cascade models are incorrectly included.
- **[science]** Figure 6: The x-axis lacks labels identifying the specific models or systems represented by the bar clusters; the caption states 'Models listed in Appendix' but does not provide the mapping needed to interpret the data.
- **[science]** Figure 6: The y-axis is labeled 'Mean Delta' but lacks a unit or metric name (e.g., 'Pass@1', 'Accuracy'), making the magnitude of the perturbation effects ambiguous.

- [**writing**] Figure 6: The caption contains a typo ‘Perturbation effect on pooled across domains’ where the specific metric name is missing (likely ‘EVA-X’ based on the filename).
- [**science**] Figure 7: The caption states ‘Models are sorted ascending,’ but the bars are sorted descending (0.971 at top to 0.592 at bottom).
- [**science**] Figure 7: The caption contains unrendered placeholders (‘aggregates the two cascades...’, ‘saturates near 1.0...’, ‘gap to the next-best STT ()’) where specific model names should be, making the text unintelligible.
- [**writing**] Figure 7: The y-axis labels are truncated (e.g., ‘Scribe-v2.2-Realtime’ is cut off at the top), reducing readability.
- [**fatal**] Figure 9: The caption contains unrendered LaTeX placeholders (‘and’, ‘mean $$ 95% CIs’) and lists specific system names (e.g., ‘+ +’) that are missing from the text, making the description of the plotted data unreadable.
- [**science**] Figure 9: The legend defines ‘Hybrid (AudioLLM + TTS)’ with a green diamond, but the caption states ‘Hybrid (n=2) is shown as individual system points only’ (referencing Figure 1’s convention), creating a contradiction between the legend’s implication of a mean point and the caption’s description of individual points.
- [**writing**] Figure 9: The caption references ‘On the plot’ and ‘On the plot’ without specifying which subplots (e.g., (a) vs (b)) correspond to these descriptions, despite the text implying a comparison of axis ranges.
- [**writing**] Figure 10 caption is truncated mid-sentence at the end of the description for panel (b) (‘shaded [threshold_sensitivity.pdf]’), cutting off the explanation of the shaded bands.
- [**writing**] Figure 10 caption contains unrendered LaTeX placeholders (‘$_t t$‘) instead of formatted symbols (e.g., t_{tt} or au_{tt}), reducing readability.
- [**science**] Figure 11: The caption claims to show ‘correlation’ (a single aggregate statistic), but the plot displays 7 individual data points representing per-system means. The figure should be described as a scatter plot of ‘Mean transcription accuracy vs Mean task completion’ rather than a ‘correlation’ plot, or the caption should clarify that the correlation coefficient (r=0.930) is the result of the analysis shown.
- [**writing**] Figure 11: The title text ‘Cascade systems: transcription vs task completion’ and the statistical summary ‘Pearson r = 0.930, p = 0.002, n = 7 systems’ are rendered as part of the image title rather than being integrated into the caption or axis labels, which is inconsistent with standard scientific figure formatting.
- [**writing**] Figure 12: The legend is illegible due to small font size and dense text, making it impossible to distinguish between the 12 system names listed.
- [**science**] Figure 12: The caption states the k=5 point is ‘identically zero’ due to a single-anchor draw, yet the plot shows non-zero error bars (vertical lines) at k=5 for several metrics, contradicting the explanation.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific acronyms and jargon that are not defined at their first point of use, creating a barrier for readers outside the immediate voice-agent subfield. While the Appendix contains a definitions section, critical terms such as **S2S** (Speech-to-Speech), **LALM** (Large Audio Language Model), **VAD** (Voice Activity Detection), and **STT** (Speech-to-Text) appear in the Introduction and Methodology without immediate explanation. For instance, Section 1 mentions "cascade, hybrid, S2S" without clarifying what S2S entails for a general audience.

Furthermore, the paper introduces custom metrics like **pass@k** and **pass^k** (reliability) in the Introduction and Results without defining the notation or the underlying logic before presenting the data. The term **IAA** (Inter-Annotator Agreement) is used in Section 4.2 without expansion. Additionally, the phrase "**barge-in**" in Section 3.2 is industry jargon that should be replaced with a plain description (e.g., "interrupting the agent") to ensure clarity.

To improve accessibility, the authors should define all acronyms and specialized metrics at their first occurrence in the main text, rather than relying on the reader to cross-reference the Appendix. This includes expanding "S2S," "LALM," "VAD," "STT," "IAA," and the custom pass-metrics immediately upon introduction.

Action items.

- [**writing**] Define 'S2S' (Speech-to-Speech) and 'LALM' (Large Audio Language Model) at their first occurrence in the Introduction or Related Work, rather than deferring definitions to the Appendix.
- [**writing**] Replace the acronym 'IAA' (Inter-Annotator Agreement) with the full term or define it immediately upon first use in Section 4.2.
- [**writing**] Define 'VAD' (Voice Activity Detection) and 'STT' (Speech-to-Text) at first use in the Introduction or Methodology, as these are not universally known to non-specialist readers.
- [**writing**] Replace the term 'barge-in' in Section 3.2 with a plain English description (e.g., 'interrupting the agent') to improve accessibility for general readers.
- [**writing**] Define 'pass@k' and 'pass^k' (reliability metric) explicitly in the Introduction or Methodology before using them in the results, as they are central to the paper's claims.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a logical framework for evaluating voice agents, but several conclusions do not strictly follow from the presented data or contain internal inconsistencies regarding metric definitions.

First, the Introduction claims "no system exceeds 0.5 on both EVA-A and EVA-X pass@1." However, Table 1 (Section 4.3) lists the GPT-Realtime system with EVA-A pass@1 = 0.467 and EVA-X pass@1 = 0.566. While 0.467 is below 0.5, the phrasing suggests a hard frontier where *no* system comes close, whereas the data shows a system clearing 0.45 on both. More critically, the text later states "Only GPT-Realtime (0.47, 0.57) clears 0.4 on both," which contradicts the initial "no system exceeds 0.5" claim if the intent was to highlight a performance ceiling. The logic of the "frontier" claim is muddled by the specific numbers provided.

Second, the definition of the composite "pass" metric in Section 3.2 creates a logical gap. The text states: "A conversation passes a dimension only if *all* component metrics meet thresholds (e.g., EVA-A pass: task completion=1, faithfulness >= 0.5, speech fidelity >= 0.95)." However, the results in Table 1 report "pass@1" scores (e.g., 0.207, 0.490) which are fractions of trials passing. If "speech fidelity" is a continuous score (0-1) as implied by the ">= 0.95" threshold, but the "Task Completion" is binary (0/1), the aggregation logic for the *system-level* pass rate is not fully derived. Specifically, if a system has high speech fidelity but low task completion, the pass rate drops to 0. The paper does not explicitly show the correlation between the component pass rates and the composite pass rate, making the claim that "peak and reliable performance diverge" (median gap 0.44) difficult to verify logically without seeing the distribution of the component failures.

Third, the conclusion asserts that "cascade systems lead in accuracy" while "S2S systems lead in experience." The data in Table 1 shows the highest EVA-A score for a Cascade system is 0.504 (Nova+GPT+Sonic), while the highest for S2S is 0.467 (GPT-Realtime). The difference is marginal (0.037) and within the standard error of some systems (e.g., ±0.044). Furthermore, the Cascade systems show a massive range (0.205 to 0.504), whereas S2S systems are more clustered (0.163 to 0.467). Claiming a categorical "lead" based on a single system's peak performance, while ignoring the variance and the fact that the top S2S system is statistically close to the top Cascade system, is a logical overreach. The paper should qualify this as "the best-performing cascade system outperforms the best S2S system in accuracy, but the architectures show significant overlap."

Finally, the "Robustness Analysis" claims "S2S accuracy unchanged" under accent perturbations, yet Figure 4 (perturbation-eva-a-pass-pooled) and the text in Section 4.3 show a drop for S2S systems (e.g., GPT-Realtime drops from 0.467 to ~0.42, a ~0.05 drop). The text states "S2S accuracy unchanged" but then immediately says "S2S remains within 5 points" for combined perturbations. The logic here is inconsistent: a 5-point drop is a change, not "unchanged." The conclusion should reflect that S2S is *more robust* (smaller drop) rather than "unchanged."

Action items.

- **[writing]** The paper presents a logical framework for evaluating voice agents, but several conclusions do not strictly follow from the presented data or contain internal inconsistencies regarding metric definitions. First, the Introduction claims "no system exceeds 0.5 on both EVA-A and EVA-X pass@1." However, Table 1 (Section 4.3) lists the GPT-Realtime system with EVA-A pass@1 = 0.467 and EVA-X pass@1 = 0.566. While 0.467 is below 0.5, the phrasing suggests a hard frontier where *no* system comes close,

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the performance landscape of voice agents that slightly overreach the specific data points presented in the tables.

First, the Abstract states: "no system exceeds 0.5 on both EVA-A pass@1 and EVA-X pass@1." While technically true for the specific systems listed (the highest EVA-A is 0.504, but its EVA-X is 0.007; the highest EVA-X is 0.589, but its EVA-A is 0.292), the phrasing creates a false impression of a universal "accuracy-experience frontier" where *no* system can be competent in both. The data shows systems like GPT-Realtime (0.467 EVA-A, 0.566 EVA-X) are quite balanced, just not crossing the arbitrary 0.5 threshold on both simultaneously. The claim should be softened to reflect that "no system achieves *simultaneous high performance* (e.g., >0.5) on both metrics" to avoid implying a fundamental impossibility rather than a current benchmark result.

Second, the Conclusion asserts: "S2S systems lead in experience... while cascade systems lead in accuracy." This is an over-generalization of the architectural comparison. Table 1 shows significant variance within architectures. For instance, the "Scribe + GeminiFlash + Conversational" cascade system scores 0.490 on EVA-A, which is higher than the "Ultravox" hybrid (0.270) and the "GPT-Realtime-Mini" S2S system (0.163). Conversely, the "GPT-Realtime" S2S system scores 0.566 on EVA-X, but the "Whisper + Qwen + Voxtral" cascade scores 0.684. The data supports a trend, not a strict rule. The conclusion should be qualified to state that S2S architectures *tend* to excel in experience metrics while *certain* cascade configurations excel in accuracy, rather than presenting it as a definitive architectural law.

Finally, the Abstract cites a "mean Δ up to 0.314" for robustness gaps. The value 0.314 is present in the data (likely a specific worst-case drop in task completion for a cascade system under combined perturbations), but presenting it as a "mean Δ" without qualification in the abstract is misleading. If it represents a maximum observed drop, the text should say "worst-case drop" or "maximum observed degradation." Using "mean" implies an average across all systems and conditions, which the data does not support (most drops are smaller). This distinction is crucial for understanding the severity of the robustness gaps.

Action items.

- **[writing]** The Abstract claim 'no system exceeds 0.5 on both EVA-A and EVA-X' is technically true but misleadingly absolute. It implies a hard frontier, whereas GPT-Realtime (0.467/0.566) is near-balanced. Rephrase to 'no system achieves simultaneous high performance (>0.5) on both metrics' to avoid implying a fundamental impossibility.

- [**writing**] The Conclusion states 'S2S systems lead in experience... while cascade systems lead in accuracy.' This overgeneralizes Table 1, where specific cascades (e.g., Scribe+GeminiFlash) outperform some S2S/Hybrid systems in accuracy, and some cascades outperform S2S in experience. Qualify the claim to reflect trends rather than strict architectural dichotomies.
- [**writing**] The Abstract cites a 'mean Δ up to 0.314' for robustness gaps. This value likely represents a worst-case drop, not a mean. Using 'mean' misleads readers about the average impact. Clarify as 'worst-case drop' or specify the exact metric and condition to ensure accurate interpretation of the robustness gaps.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a comprehensive framework for evaluating voice agents but falls short in explicitly addressing safety and ethics risks, which is critical given the high-stakes domains (Healthcare, Airline) evaluated.

Primary Concern: Lack of Safety Metrics and Results In the "Limitations" section (Section 6), the authors state: "No assessment of harmful outputs or PII exposure." While this is an honest admission, it is insufficient for a paper claiming to evaluate enterprise-grade voice agents. The "Ethics Statement" (Section 6) further notes that the framework "cannot guarantee models won't generate harmful output."

- **Issue:** The paper evaluates 12,780 conversations across 12 systems but provides zero data on safety failures (e.g., hallucinated medical advice, unauthorized data access, or PII leakage).
- **Requirement:** The authors must either (a) include a "Safety" dimension in their EVA-A/EVA-X metrics with reported failure rates, or (b) provide a dedicated "Safety Analysis" section in the results (Section 4) quantifying the frequency of harmful outputs observed during the trials, even if the framework did not automatically flag them. Relying solely on "content filtering is recommended" (Ethics Statement) is not a substitute for empirical safety evaluation in a benchmark paper.

Secondary Concern: Bias and Fairness in Perturbations The "Limitations" section admits that robustness testing covers only "one accent (French) and one noise environment."

- **Issue:** This limited scope may mask safety-critical failures. For instance, a voice agent might hallucinate medical instructions more frequently for non-native speakers or under specific noise conditions not tested.

- **Requirement:** The “Ethics Statement” should be expanded to explicitly discuss the *safety implications* of this limited accent/noise coverage, rather than treating it purely as a performance limitation.

Tertiary Concern: Auditability The “Reproducibility Statement” notes that full reproduction requires commercial API access.

- **Issue:** This restricts the ability of the broader community to independently audit the framework for safety vulnerabilities or to stress-test the models for harmful outputs without incurring significant costs.
- **Requirement:** Clarify if the scenario generation pipeline (SyGra) and the evaluation logic are fully open-source to enable third-party safety auditing, even if the model inference itself requires paid APIs.

The paper is scientifically sound in its methodology but requires a more robust treatment of safety and ethics to be suitable for publication in a venue concerned with responsible AI.
Action items.

- [**science**] The Ethics Statement (Section 6) explicitly admits the framework ‘cannot guarantee models won’t generate harmful output’ and recommends content filtering. However, the paper lacks a dedicated ‘Safety Evaluation’ section or results quantifying the rate of harmful, biased, or PII-leaking outputs observed during the 12,780 trials. Given the high-stakes domains (Healthcare, Airline), this omission is a significant gap in the safety assessment.
- [**writing**] The ‘Limitations’ section notes that scenarios are in English and use only one French accent. This creates a potential fairness/bias risk where the framework may fail to detect safety failures (e.g., hallucinated medical advice) in underrepresented accents or languages. The authors should explicitly discuss this limitation as a safety concern, not just a performance one.
- [**writing**] The ‘Reproducibility Statement’ notes that full reproduction requires commercial API access. While acceptable for science, this creates a barrier for independent safety auditing of the framework’s ability to detect harmful outputs. The authors should clarify if the synthetic data generation pipeline (SyGra) is fully open-sourced to allow third-party safety stress-testing without API costs.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a comprehensive evaluation framework with a large sample size (213 scenarios, 12 systems, ~12,780 trials), which provides a strong foundation for the central claims. The use of deterministic metrics (Task Completion via SHA-256) and rigorous LLM-as-Judge validation (IAA κ 0.777–0.845) strengthens the evidence for the reported accuracy and experience scores. The variance decomposition analysis (Sec 4.2) correctly identifies trial stochasticity as the primary noise source, justifying the multi-trial approach.

However, the robustness of the central claims regarding architectural differences and perturbation effects is weakened by specific experimental design choices. First, the robustness analysis (Sec 4.3) relies on a single French accent voice and a single noise environment (coffee shop). While the results show significant drops, the claim that "accent hits cascade accuracy" is potentially confounded by the specific acoustic characteristics of the chosen voice and noise profile rather than the general category of "accent." The Limitations section acknowledges this, but the main text presents the findings as general architectural traits without sufficient qualification.

Second, the "reliability" metric (pass^k) assumes independent trials to model the probability of consistent success. The paper notes that trial stochasticity is high, but does not explicitly detail the randomization seeds or temperature settings used to ensure true independence between the $k = 5$ trials per scenario. If trials are correlated (e.g., same seed), the calculated reliability gap (median 0.44) may be inflated or deflated.

Finally, the claim of a "median gap of 0.44" between peak (pass@k) and reliable (pass^k) performance is a strong statistical assertion. The paper reports this as a single number without providing the distribution (e.g., interquartile range) or a visual representation (e.g., boxplot) of the gaps across systems. Given the high variance in trial outcomes, understanding the spread of this gap is crucial to assessing whether it is a consistent phenomenon or driven by a few outlier systems.

Action items.

- [**science**] Clarify the statistical basis for the 'median gap of 0.44' claim (Sec 1, Sec 4.3). The text states this is a median gap between pass@k and pass^k, but the calculation method (per-scenario vs. per-system) and the distribution of these gaps are not described. Provide the interquartile range or a histogram to demonstrate robustness against outliers.
- [**science**] Address the confounding variable in the robustness analysis (Sec 4.3, Limitations). The study uses only one specific French voice and one coffee shop noise profile. The claim that 'accent hits cascade accuracy' may reflect specific acoustic properties of that voice/noise pair rather than general accent robustness. Explicitly state this limitation as a threat to external validity or add a sensitivity analysis if data permits.
- [**science**] Justify the 'pass^k' reliability metric (Sec 2.2, App Defs). The metric assumes independent trials to calculate the probability of passing all k attempts. However, the text notes 'Trial stochasticity is the dominant variance source' (Sec 4.2). If trials share the same underlying model state or prompt seed, independence is violated. Confirm the randomization strategy (e.g., temperature, seed) used to ensure trial independence.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in EVA-Bench is ambitious but lacks necessary rigor in reporting and inference. While the sample size of trials (~12,780) is large, the unit of analysis for many claims is the system (n=12) or architecture class (n=3), which severely limits statistical power for comparative claims.

First, the reporting of uncertainty is inconsistent. Table 1 presents metrics as "Mean ± SD" (e.g., 0.207 ± 0.041), yet the text and Figure captions describe these as "95% percentile bootstrap intervals." Standard deviation and confidence intervals are distinct statistical concepts; conflating them misrepresents the precision of the estimates. The authors must either relabel the table columns as "Mean (SD)" or recompute the values as 95% CIs and update the table accordingly.

Second, the paper performs extensive hypothesis testing without addressing the multiple comparisons problem. With 12 systems evaluated across multiple metrics (EVA-A, EVA-X, and sub-metrics), the number of pairwise comparisons is substantial. The claim that "81% [of turn-taking metrics] show significant degradation" (Section 4.3) implies a large number of t-tests or ANOVAs were run. Without correction (e.g., Bonferroni, Benjamini-Hochberg), the reported p-values (e.g., $p < 0.0001$ in Section 4.2) are likely inflated. The authors should apply a correction method and report adjusted p-values, or explicitly state that the analysis is exploratory.

Third, the correlation analysis in Section 4.4 (Pearson $r = 0.93$, $p = 0.002$) is based on only 7 data points (the cascade systems). With n=7, the degrees of freedom are too low to robustly support a parametric correlation claim, especially given the potential for outliers. The authors should report the exact test used, consider a non-parametric alternative like Spearman's rank correlation, and acknowledge the limited power of this specific analysis.

Finally, the variance decomposition claim ("Trial stochasticity is the dominant variance source... $p < 0.0001$") lacks methodological detail. The specific statistical test (e.g., F-test for variances, Levene's test) and the model used to partition variance (e.g., mixed-effects model) are not described. Given the binary nature of "pass" metrics, standard ANOVA assumptions may be violated. A brief description of the statistical model and assumptions is required to validate this claim.

Action items.

- **[science]** The paper reports p-values (e.g., $p < 0.0001$ for variance decomposition) but does not specify the statistical test used (e.g., Levene's, F-test) or the distributional assumptions. Given the non-normal nature of pass/fail metrics, clarify the test statistic and validity of the p-values.
- **[science]** Multiple comparisons are performed across 12 systems and 6+ metrics without

correction (e.g., Bonferroni, Holm). With ~72+ pairwise tests implied, the risk of Type I error is high. Report adjusted p-values or justify the lack of correction.

- [**science**] Confidence intervals are reported as '±' standard deviations (e.g., 0.207 ± 0.041) in Table 1, but the text and figure captions refer to '95% percentile bootstrap intervals'. Standard deviation is not a confidence interval. Explicitly label these as SD or recompute as 95% CIs for consistency.
- [**science**] The correlation claim (Pearson r = 0.93, p = 0.002) between transcription accuracy and task completion is based on n=7 systems (cascade only). This sample size is insufficient for robust correlation inference. Report the exact test used and consider non-parametric alternatives (Spearman) given the small N.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high level of technical sophistication, but the writing quality occasionally suffers from over-reliance on LaTeX macros and dense sentence structures that impede immediate readability for a general audience.

The most significant issue is the frequent use of undefined or opaque LaTeX macros (e.g., `\metricbehavior`, `\passatone`, `\framework`) directly within the prose of the main text (Section 3.1, Section 4.2). While these are defined in the appendix, their presence in the main narrative forces the reader to constantly cross-reference or guess the meaning, breaking the flow. For instance, in Section 3.1, "Automated checks for \metricbehavior~and \metricuserspeechfidelity" should be written out as "Automated checks for User Behavioral Fidelity and User Speech Fidelity" to ensure the text stands alone.

Additionally, some statistical comparisons in Section 4.2 are presented with ambiguous phrasing. The statement "S2S systems dominate EVA-X (mean turn-taking 0.82–0.83 vs. cascade 0.28–0.58)" does not explicitly state that these are aggregate means across the respective architecture classes. While likely clear to an expert, a more explicit phrasing (e.g., "S2S systems, with a mean turn-taking score of 0.82–0.83, dominate...") would improve clarity.

Finally, the Limitations section contains a slightly confusing parenthetical regarding judge bias: "(e.g., GPT-5.4 evaluated with GPT-5.2 judges)". This phrasing is ambiguous; it is unclear if this refers to a version mismatch causing bias or a specific experimental condition. Clarifying the relationship between the evaluated model and the judge model in this example would prevent misinterpretation.

Overall, the paper is well-structured, but smoothing out these macro-heavy sentences and clarifying statistical comparisons will significantly enhance its accessibility.

Action items.

- **[writing]** In Section 3.1 (Conversation Simulation), the phrase 'Automated checks for \metricbehavior~and \metricuserspeechfidelity' uses undefined LaTeX macros. Replace these with the full metric names (e.g., 'User Behavioral Fidelity' and 'User Speech Fidelity') or ensure the definitions are visible in the main text flow to avoid reader confusion.
- **[writing]** In Section 4.2 (Main Findings), the sentence 'S2S systems dominate EVA-X (mean turn-taking 0.82–0.83 vs. cascade 0.28–0.58)' lacks a clear subject for the comparison. Clarify whether these ranges refer to the mean scores across all S2S systems versus all cascade systems, or specific subsets, to ensure the statistical claim is unambiguous.
- **[writing]** In the Limitations section, the phrase 'LLM/LALM judges may exhibit stylistic biases (e.g., GPT-5.4 evaluated with GPT-5.2 judges)' is slightly confusing. Clarify if this refers to a version mismatch between the model being evaluated and the judge model, or a specific experimental setup, to prevent misinterpretation of the bias source.